ONE FLAT FIELD COVER MAKES ALL FIELD POINTS FLAT AND SURJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2626-2632 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2634-2651, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 29 of queue_stacks-properties.json (pb=588, stmt 2626-2632, proof 2634-2651).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-flat-cover-by-field' also exists in stacks-decent-spaces — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma section-properties-morphisms: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 378–401. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05VY: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2651–2651. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{Z}$ be an algebraic stack. Let k be a field and let* $\mathrm{Spec}(k) \to \mathcal{Z}$ *be surjective and flat. Then any morphism* $\mathrm{Spec}(k') \to \mathcal{Z}$ *where k' is a field is surjective and flat.*

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Consider the fibre square

$$\begin{array}{ccc} T & \longrightarrow & \mathrm{Spec}(k) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k') & \longrightarrow & \mathcal{Z} \end{array}$$

Note that $T \to \mathrm{Spec}(k')$ is flat and surjective hence T is not empty. On the other hand $T \to \mathrm{Spec}(k)$ is flat as k is a field. Hence $T \to \mathcal{Z}$ is flat and surjective. It follows from Morphisms of Spaces, Lemma 05VY (via the discussion in Section section-properties-morphisms) that $\mathrm{Spec}(k') \to \mathcal{Z}$ is flat. It is clear that it is surjective as by assumption $|\mathcal{Z}|$ is a singleton. $\square$